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Students' perceptions of multiliteracies development using AI-assisted portfolio assessment

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ABSTRACT

In higher education, Generative Artificial Intelligence (GenAI) has opened new possibilities for developing student multiliteracies and set new requirements for multiliteracies pedagogies. While research has investigated GenAI's affordances and limitations in language teaching, empirical studies on integrating GenAI in formative assessment practices to empower students' multimodal competence development remain lacking. This study examines how students develop multiliteracies throughout e-portfolio compilation in an EFL, GenAI-supported portfolio assessment classroom and uncovers students' perceptions of using GenAI in formative assessment. Employing a qualitative design over 16 weeks, 50 freshmen students were divided into groups to collaboratively create WordPress-based portfolios featuring multimodal writing about touring routes recommendations. Students reflected and revised their multimodal artefacts based on self, peer, teacher and/or AI-generated feedback. Students' reflective journals, focus group interviews and narrative inquiries were collected and analysed thematically. Findings show students develop multiliteracies through three stages, moderate transformation from passive to proactive learning, growing confidence and multimodal awareness, enhanced metacognitive development and professional growth. Students' perceived affordances and challenges are discussed. This study contributes to portfolio assessment scholarship by introducing GenAI as a key component in assisting students' feedback and assessment practices. Pedagogical implications are proposed for optimizing GenAI use to benefit students' learning.

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multisource feedback;
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1. Introduction

Since the New London Group (1996) coined the idea of multiliteracies, there has been growing recognition that literacy encompasses multisituational and multiform (Cope & Kalantzis, 2023) of meaning-making shaped by technological advancement. In an age of growing digitalization, multiliteracies pedagogies have played a crucial role in aiding students' multimodal knowledge construction and sense-making. In English as a Foreign Language (EFL) contexts, the concept of

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multiliteracies has triggered a positive change in literacy education, and teachers have implemented various pedagogical approaches to foster students' multiliteracies development (S.-Y. Lee et al., 2021; Mirhosseini & Emadi, 2022).

The emergence of Generative Artificial Intelligence (GenAI) has transformed the language education landscape, with research corroborating the effectiveness of GenAI-integrated teaching and learning in EFL contexts (Dai & Liu, 2024; Ma et al., 2024; Meniado et al., 2024). This transformation has further accelerated the evolution of literacy practices in EFL classrooms. Kalantzis and Cope (2025, p. 2) propose a new paradigm of "cyber-social literacy learning", which conceptualizes AI not as a replacement of human writing ability, but as a co-partner that provides "scaffolded, dialectical feedback" to support learner users in a virtual environment.

Against this backdrop, portfolio assessment, as a manifestation of formative assessment, has gained significant momentum in the digital age owing to its intertextual and multimodal nature. In EFL contexts, the creation and curation of multimodal artefacts and the exchange of multimedia feedback (Lam, 2023) align closely with the paradigm of multiliteracies pedagogies. When used with GenAI, particularly with timely feedback, e-portfolios offer promising potential for supporting students' literacy development while students retain agency over their learning journey and creative multimodal expressions. Existing research has extensively examined the affordances and limitations of GenAI in language teaching and learning, primarily focusing on linguistic aspects (e.g. improved argumentation skills in K. Guo et al., 2023). There remains a lacuna in empirical studies investigating how GenAI-assisted formative assessment practices can influence students' multimodal competence development.

Therefore, this study examines how students develop multiliteracies through a GenAI-assisted portfolio assessment project in an EFL classroom at a Chinese university. It seeks to understand students' perceptions of GenAI assistance in multimodal knowledge construction in formative assessment practices. The study employs a qualitative research design and documents students' multiliteracies development over a 16-week semester, during which participants engaged in a GenAI-assisted portfolio compilation process supported by hands-on workshops, provision of synchronized feedback and opportunities for learning-oriented assessment practices. Through the collection and analysis of reflective journal entries, focus group interviews, and narrative inquiries, this study explores students' developmental trajectories of multiliteracies.

This research contributes to the formative assessment scholarship empirically by integrating GenAI tools in feedback practices and thus making GenAI-assisted portfolio assessment a transformative literacy practice in EFL contexts. The study examines how students reflect on multisource feedback (self, peer, teacher, and AI-generated ones) in their multimodal meaning-making and portfolio compilation, offering valuable insights for understanding multiliteracies education in the GenAI era. The findings are likely to have significant implications for teachers who seek to integrate GenAI into formative assessment practices while cultivating students' multiliteracies development as self-regulated learners.

2. Literature review

2.1. *Multiliteracies in EFL context*

Traditional language education has long prioritized paper-based, text-centric literacy practices, which treat reading and writing as isolated, standardized skills (Kalantzis & Cope, 2023). However, the digitalization of communication has triggered a paradigm shift. Specifically, literacy in the 21st century must be reconceptualized as multiliteracies in its plural to reflect the dynamic, multimodal, and technological diversity of a globalized world (Cope & Kalantzis, 2023). This perspective advocates that effective communication requires learners to master not only textual but also visual, spatial, auditory, and gestural semiotic resources. Research also explores embodied (Lim, 2020) and sensory (Mills, 2016) dimensions of meaning-making, which suggests that literacy involves the whole person and multiple sensory channels.

GenAI has further transformed the literacy education landscape. Kalantzis and Cope (2025) coined the term “cyber-social literacy learning”, in which AI serves as an online collaborator providing scaffolded, dialectical feedback for user writers instead of replacing human creativity and writing ability. This new paradigm enables learners to engage critically with AI feedback in dynamic and dialogic interaction. It helps maintain learner agency while leveraging technological support for multimodal meaning-making.

Recent multiliteracies research in the EFL context reveals several key developments, with digital multimodal composing (DMC) central to this transformation. In language classrooms, students engage in creating digital videos, infographics, podcasts, and other forms with multiple modes and multimodal ensembles (Hafner, 2014; Hafner & Miller, 2011). Towndrow and Pereira (2018) further extend it by showing the development of learners’ “semiotic awareness” through digital storytelling, which refers to the ability to select appropriate modes for a specific purpose. It echoes with what Kress (2010, p. 157) termed as “semiotic harmony”, demonstrating through the orchestration, that is the assembly and design, of different modes into coherent ensembles where each mode contributes its specific affordances to create unified meaning. Hafner and Ho (2020) further address DMC assessment challenges through a process-based model which considers both multimodal affordances and orchestration (“semiotic harmony”). It succinctly shows how formative assessment supports students’ multimodal competence development.

These multimodal practices have also transformed pedagogy and student engagement. Empirical studies highlight the transformative potential of multiliteracies pedagogies in enhancing learners’ meaning-making and autonomous learning (S.-Y. Lee et al., 2021; Mirhosseini & Emadi, 2022). The integration of multimodal resources, including digital video making (Yeh, 2018), PowerPoint presentations (Z. Zhang, 2013), and blended learning environments (D. Zhang & Zou, 2020), has proven effective in nurturing students’ multiliteracies awareness and engagement with synthesizing various semiotic resources. These findings emphasize the importance of apt selection (Lim et al., 2022), which means choosing the most effective semiotic modes for specific communicative purposes.

In multiliteracies pedagogies, explicit multimodal literacy instruction is achieved by promoting students’ metalanguage development, a way that enables students to describe and discuss their semiotic choices (Lim et al., 2022). This metalinguistic

awareness has emerged as valuable in connecting classroom learning with out-of-school literacy practices (W. J. Liang & Lim, 2021) and facilitating the metacognitive feedback process in digital multimodal composing assessment (Liu & Lam, 2025). While these studies provide valuable insights into multiliteracies practices, there remains a lacuna in understanding how students' multiliteracies develop.

Owing to the advent of GenAI and a cyber-social literacy learning paradigm, AI-supported portfolio assessment offers a promising approach for multiliteracies development with dialogic feedback from multiple sources.

2.2. Portfolio assessment and self-regulated learning

Writing portfolio assessment, particularly in its digital form, has emerged as a powerful pedagogical tool that bridges assessment and learning in language education. According to Yancey (2019), e-portfolios are digital dossiers for learners' multimodal artefacts, which record the creation, curation and reflection along their language learning journeys. The value of portfolio assessment in developing students' multiliteracies is threefold. First, it is an effective pedagogy that enhances learner agency (Lam, 2024a), promotes self-efficacy (H. Guo & Li, 2024), and stimulates students' deep reflection (Hamp-Lyons & Condon, 2000). Second, it serves as a facilitator of formative assessment practices (I. Lee, 2017). More importantly, it incorporates a process that enables students to integrate various semiotic resources to construct and communicate meaning (Lam, 2024a).

During compilation, e-portfolios provide a feedback-rich learning environment where students engage in self-assessment, peer feedback, and teacher evaluation while continuously refining their work (Yancey, 2019). This multi-layered feedback paradigm mirrors what Carless (2016, pp. 2–4) terms a “dialogic” feedback environment, which operates through integrated guidance cycles, peer feedback, technology-facilitated feedback, internal feedback, and dialogic written feedback. As part of the formative assessment process, students will be given feedback along with their portfolio compilation and asked to act upon it. They will then use this input to make adjustments and improvements to their works-in-progress (Lam & Gan, 2024). Therefore, with feedback and reflection as a cornerstone, research has highlighted the potential of e-portfolios in fostering self-regulated learning (e.g. Lam, 2022; C. Liang et al., 2015). Tran and Qing (2024) demonstrate how online peer feedback enhances self-regulated writing strategies, while Chang and Sun (2024) emphasize immediate AI-generated feedback in supporting personalized learning paths.

However, challenges exist in implementing effective feedback practices. Traditional teacher feedback often functions more as directives than dialogic exchanges, where teachers rarely follow up on students' revisions (Carless, 2006). Additionally, students sometimes struggle to comprehend teacher feedback or receive it too late to make meaningful revisions (Gan & Lam, 2020). From the teachers' side, feedback provision in formative assessment sometimes puts a strain on already heavy workloads (Lam, 2018), and some teachers might emphasize more on error feedback rather than formative feedback that motivates students' autonomous learning (I. Lee, 2014). These limitations denote the need for more dynamic and timely feedback mechanisms. The integration of GenAI tools in portfolio assessment presents new opportunities for feedback giving. Although there are concerns about information overload in

AI feedback (Chen et al., 2024) and its ethical use (Meniado et al., 2024), e-portfolios are likely to offer unique advantages in validating and authenticating student work and authorship through timestamped feedback trails and evidence-referenced reflection (Lam & Gan, 2024).

GenAI-assisted portfolio assessment emerges as a particularly promising approach against the backdrop of cyber-social literacy learning as it positions AI as a collaborative partner that provides timely and scaffolded feedback while maintaining active learner agency throughout the portfolio compilation process.

2.3. GenAI and education assessment

GenAI's integration into educational assessment offers both opportunities and challenges for language learning. Research highlights its ability to support personalized feedback and multimodal composing (Ma et al., 2024; Meniado et al., 2024), and how language teachers can better utilize GenAI to address problems and empower their teaching (Tafazoli, 2024). In formative assessment, GenAI shows promise in enhancing feedback quality (Gardner et al., 2021) and fostering self-regulated learning (Hopfenbeck et al., 2023). However, besides the concern over academic integrity, over-reliance on GenAI is prevalent and more likely to undermine critical thinking and learner autonomy, particularly when students use them uncritically in formative assessment contexts (Meniado, 2023). Therefore, Xia et al. (2024) call for pedagogical scaffolding that both nurtures students' goal-setting strategies and cultivates their critical thinking skills. ChatGPT literacy training focusing on prompt refinement is also deemed essential in maximizing GenAI affordances (Ma et al., 2024).

Although studies have examined GenAI's role in summative assessment (Meniado, 2023), research on its integration into formative assessment practices remained limited. Memarian and Doleck (2024) call for research investigating how students process and apply AI-generated formative feedback. Understanding students' use of GenAI in formative assessment is crucial for sharpening pedagogies that guide learners in balancing optimal AI use and autonomous learning capacity. Education inclusiveness is also maintained as structured support in formative assessment can promote equitable access for students with multiple ethnic backgrounds and nuanced academic profiles (Tafazoli, 2024).

The literature reveals the need for empirical studies that examine the developmental trajectory of students' multiliteracies through GenAI integration in formative assessment practices. The current study addresses this gap by investigating how university-level students engage with GenAI-supported portfolio assessment and develop their multiliteracies through this compilation process. The study was guided by the following two research questions.

- (1) How are EFL students' multiliteracies developed in AI-assisted portfolio assessment?
- (2) What are EFL students' perceptions of the affordances and challenges of AI use in portfolio compilation?

3. Methodology

3.1. *Research site and informants*

The current study was conducted in an application-oriented university in Southeast China that emphasizes cultivating students' professional skills and competencies and bridging the gap between theories and practices in real social contexts. The university encourages teachers' innovations in teaching practices and formative assessment to foster students' competencies while placing less emphasis on the high-stakes testing culture and summative results, enabling the smooth implementation of this innovative multiliteracies pedagogy with institutional support (Lam, 2020b).

The participants for the research were 50 first-year Tourism Management major students in a pre-assigned class. All the students had taken the placement tests (listening, reading comprehension, and writing adapted from College English Test Band 4) in their first week, and they demonstrated a variety of proficiency in English equivalent to the CEFR A1-B2 level. Albeit variance in students' initial language proficiency, the naturalist setting of the research makes it possible for the multiliteracies pedagogy to accommodate a wider context.

Prior to the study, students had limited or no experience and knowledge with AI tools since they had just finished the Chinese College Entrance Examination and graduated from mainstream public-funded high schools without knowing how to utilize AI tools for academic purposes. Most students had basic familiarity with digital content creation through mobile applications but lacked training in computer-assisted multimodal composing or portfolio compilation.

In the study, the first author also served as the instructor. The instructor-researcher was proficient with GenAI tools for education purposes and had prior experience in integrating AI feedback into writing instructions. This researcher-as-instructor paradigm enabled direct observation of students' engagement and facilitated a deeper understanding of their developmental trajectories through sustained interaction and documentation. Data triangulation with multiple sources, constant peer review, and ongoing reflective journaling are carried out to mitigate the potential researchers' bias.

3.2. *Research design*

Drawing upon the interpretivist stance, the research adopts a qualitative design to uncover the developmental trajectory of students' multiliteracies and examine the effectiveness of GenAI-assisted portfolio assessment as a multiliteracies pedagogy in an EFL context. The interpretivist approach allows for an in-depth exploration of students' nuanced perceptions and experiences in the meaning-making and compilation processes. The reason for our special focus on students' compilation experience is that it requires knowledge and skills in manipulating multimodal artefacts (creation, curation, revision, and reflection). These processes involve multimodal competence alongside formative assessment literacy, guiding students to use multisource and multimodal feedback for self- and co-regulated learning more proficiently. A triangulated design incorporating multiple data sources will likely enhance the credibility and trustworthiness of the results.

3.3. *E-Portfolio procedures and GenAI integration*

The research took place during a 16-week semester and adopted convenience sampling. 50 non-English major freshman students in a pre-assigned class were divided into 11 groups and asked to collaboratively design and recommend a touring route for a specific mainland Chinese province to foreign tourists. This aligns with the situated practices that the New London Group (1996) calls for, which involve students' personalized experiences that can augment learners' agency.

Students created multimodal artefacts at two levels. Each group was required to set up a web-based portfolio using WordPress, where every group member independently posted an expository essay describing the tourist sites and an argumentative essay providing reasons for their recommendations. These individual contributions were then synthesized into a cohesive touring route in a PowerPoint-based group portfolio, which they showcased in the in-class presentation. This group-based portfolio design was intended to provide students with a collaborative, multimodal, feedback-rich meaning-making environment where they could develop both individual multimodal composing skills and collective knowledge construction competence through negotiating meanings across diverse perspectives.

As shown in Figure 1, the e-portfolio compilation started in the first class, where the instructor illustrated the procedures in the e-portfolio project and assigned the tasks to the students with a timeline. The teacher presented one web-based portfolio and one slide-based portfolio in front of the class to demonstrate how she crafted those all-round assessment rubrics. Besides the teacher input, students were then encouraged to suggest additional perspectives and further fine-tune the rubrics centred around content development, language use, multimodal elements, and overall meaning-making coherence (see Appendix). This co-constructed rubric (Yan & Carless, 2022) is likely to guide students' self- and peer-assessment, feedback provision, and better facilitate their prompting for AI-generated feedback. Unlike traditional formative assessment that primarily targets linguistic accuracy, this rubric encompassed multimodal design choices and multimodal orchestration (Hafner & Ho, 2020), which reflects the expanded scope of assessment in multiliteracies contexts. After that, students began working in groups, planning and creating their multimodal artefacts. While the teacher recommended ERNIE-BOT and Kimi as AI feedback providers, which are widely used GenAI tools in China, students were still free to choose any GenAI tools they had access to. Then they evaluated, compared and reflected on self, peer, teacher, and AI feedback, revising both genres of writing through three writing cycles, each of which lasted for a week.

The four-phase compilation process is guided by multiple feedback mechanisms, with ongoing self- and peer assessment at every stage, while peer and teacher feedback specifically enhance the reflection and revision phases. The integration of GenAI tools followed the portfolio process shown in Figure 1, with AI serving as both a direct feedback provider and an assisted feedback facilitator. Specifically, GenAI tools provided immediate feedback on language use, content organization, and multimodal integration suggestions in the creation and curation phases to boost learning efficiency. In reflection and revision, GenAI tools were used to enhance learning outcomes and students' engagement through assisting students' uptake of teacher feedback, moderating peer discussions, and synthesizing multisource feedback. The compilation experience and multisource feedback

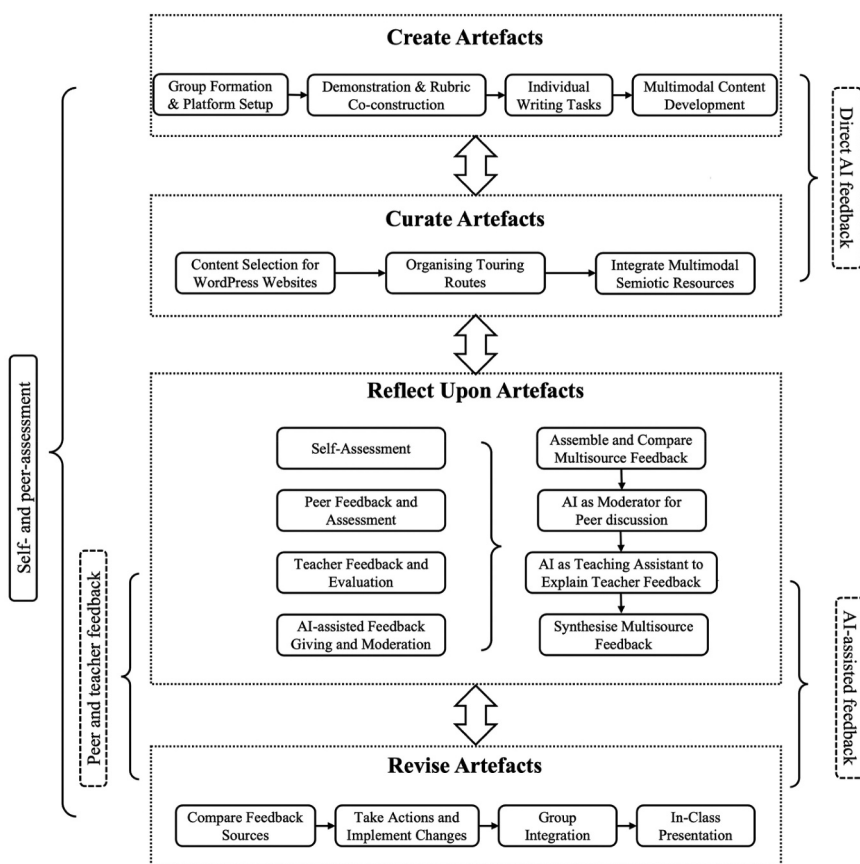


Figure 1. E-Portfolio process and GenAI integration, adapted from Lam (Lam, 2024b, p. 412).

integration are likely to develop both students' portfolio assessment literacy and self-regulated learning through authentic, situated practices in a naturalistic setting.

3.4. Data collection

After the compilation of e-portfolios, students composed their reflective journal entries in Chinese based on the guidelines that helped them critically evaluate their experiences, actions, and learning evidence since reflective journals are likely to unveil the compelling account of the participants' lived experiences. The reflective guidelines encompass a description of learning and GenAI-assisted portfolio compilation experiences, namely specific learning episodes, critical analysis of the reasons, causes and consequences of the learning experience, assessment of one's actions, and development of future actions. Fifty pieces of reflective journals were then collected. At the end of the semester, four groups of students voluntarily participated in face-to-face focus group interviews that allowed them to fully elaborate on their ideas in a less stressful environment. Questions were raised about: 1) their e-portfolio compilation experience, 2) the development of assessment and feedback skills, 3) multimodal knowledge construction on digital platforms, and 4) interactions with GenAI. The focus group interviews, each lasting for around

one hour, were conducted in the informants’ mother tongue Mandarin, and recorded upon participants’ written consent.

After the focus group, six student informants were invited to participate in the narrative inquiries based on their willingness to share and potential to provide rich content. Narrative inquiry was employed for its exploratory nature in a naturalistic setting (Barkhuizen, 2019), which aligns with the developmental trajectory of students’ multi-literacies development this study intended to investigate. As the e-portfolio compilation itself represents a developmental journey, narrative inquiry has the potential to capture this progression through students’ own voices. Additionally, this approach integrated seamlessly with coursework and as part of e-portfolios, serving dual purposes as both reflecting activities and a research instrument. The narrative inquiry was based on the narrative frames provided by the researcher, and student participants narrated their past, present, and future stories (Lam, 2020a), offering invaluable insider perspectives for understanding how their multiliteracies are shaped and developed. The narrative inquiries captured students’ learning progression from past English learning experiences to current multimodal competencies, their experiences and emotional responses during collaborative activities such as peer discussions, multimodal composing, group presentations, and their metacognitive reflections on the learning process and future aspirations. The narrative frames provided guidance and support for informants both in structure and content of what was to be delivered. Each narrative inquiry session lasted for 30–45 minutes. Furthermore, students’ multimodal artefacts from both individual and group portfolios were collected as supplementary data to cross-reference and triangulate findings from verbal data. Table 1 shows the details of the data collected and how they will be identified and addressed in the findings section.

3.5. Data analysis

The study employed a qualitative thematic analysis to explore students’ developmental trajectories of multiliteracies and perceptions of GenAI-assisted portfolio assessment. Data from reflective journal entries (*n* = 50), focus group interviews (4 groups), and narrative inquiries (6 student informants) were analysed iteratively and inductively to identify themes and patterns. Students’ multimodal artefacts were analysed through multimodal analysis, which focused on semiotic integration, evidence of improvement through feedback cycles, the reference to students’ design choices and feedback uptake documented in reflective entries, and the creative use of multimodal resources for knowledge construction. This analysis complemented verbal and self-reported data by providing tangible evidence of multiliteracies development and visualizing students’ feedback uptake and applications in their revisions.

Table 1. Data collection wrap-up.

Instruments	Number of Participants/Items	Identifications
Reflective entries	50 pieces	Student 1 to Student 50
Focus Group Interviews	4 groups	Group 1: FG1-S1 to FG1-S5 Group 2: FG2-S1 to FG2-S4 Group 3: FG3-S1 to FG2-S3 Group 4: FG4-S1 to FG4-S4
Narrative Inquiries	6 participants	Pseudonym: Lucy; Lily; James; Emily; Ella; Mike

The initial coding was done by the first author, and the results were reviewed by the second author. The thematic coding was analysed focusing on students' multimodal knowledge construction and development of multimodal competence, their e-portfolio compilation experience, the feedback and assessment experience, and their perceived benefits and challenges in GenAI assistance. All the data was analysed in Chinese to generate codes and themes, which were then translated into English by the first author and checked by the second author. Both authors are proficient in Chinese and English, guaranteeing the translated themes best represent what the informants wanted to express.

Table 2 presents the initial themes derived from the thematic analysis of fifty reflective journal entries about students' learning experiences with GenAI-assisted portfolio compilation. These themes also acted as a primary framework for the analysis of focus group interviews and narrative inquiries and were constantly revised where necessary.

3.6. Ethics

All participants provided written consent after receiving detailed information about the study's purpose, procedures, data collection methods, and their rights as participants. Owing to the dual role of instructor-researcher, students were informed that their participation in the focus group interview and narrative inquiry was voluntary and would not affect academic grades. They could withdraw from this study while continuing with the coursework. All data were anonymized, and participants' confidentiality was maintained throughout the research process.

Table 2. Initial themes from students' reflective journal entries.

Themes	Excerpts
1. Language Learning	
(a) Vocabulary ($n = 50$)	<i>My word choice has improved, and my vocabulary has expanded, especially words related to travel and tourism.</i>
(a) Grammar ($n = 50$)	<i>I've also improved in grammar. I used to write mostly simple sentences, but now I can use some subordinate clauses.</i>
2. Portfolio compilation	
(a) Design and Curation ($n = 42$)	<i>We discuss what fonts and colours to use on the website, and which areas can have images added. We'll find suitable ones to fill in.</i>
(a) Feedback giving ($n = 42$)	<i>Team members give each other feedback. Those good at English give more language suggestions, while the more artistic ones give advice on image selection and website layout.</i>
3. Multimodal competence	
(a) Multimodal awareness ($n = 31$)	<i>I will add relevant images to the PowerPoint slides and choose templates that suit our travel destinations.</i>
(a) Strategic use of semiotics ($n = 39$)	<i>Our group played soft background music during the presentation since we wanted to create a relaxing atmosphere that matches our leisure travel theme.</i>
4. GenAI assistance	
(a) Study companion ($n = 42$)	<i>AI is definitely helpful for learning, especially in terms of language. AI is like a friend who's always with me. I can ask for its help whenever I have Internet access.</i>
(a) Instant response ($n = 30$)	<i>AI can give immediate answers to my questions, and the communication with it is very smooth.</i>

4. Findings

The data analysis revealed three key themes that map the developmental trajectory of students' multiliteracies and their evolving perceptions towards GenAI-assisted portfolio assessment: (1) enhanced engagement and multiliteracies development, (2) expanded usefulness of e-portfolios for multimodal communication, and (3) challenges and pedagogical support in GenAI-assisted formative assessment. These findings address the two research questions and illustrate how iterative engagement with GenAI-assisted e-portfolio compilation and reflective practices fostered multiliteracies development while reshaping students' perceptions towards technology-mediated learning as self-regulated learners.

4.1. AI-enhanced engagement and multiliteracies development

The GenAI-assisted portfolio assessment catalysed the elevation in students' engagement, spurring the transformation from passive recipients of knowledge to reflective learners actively engaging in knowledge construction with multimodal semiotics. The themes highlighted two developmental stages: AI-empowered curation of multimodal artefacts and augmented confidence in multimodal meaning-making.

4.1.1. AI-empowered curation and redesigning of multimodal artefacts

In portfolio assessment, students are required to constantly revisit and revise the artefacts based on multisource feedback, through which students develop a deeper understanding that writing is a recursive process upon feedback and reflection. Students' engagement was deeply rooted in the cyclical process of creating, curating, reflecting, and revising the multimodal artefacts. Lucy described this shift in her narrative:

At first, I just took this project as a compulsory task. But when we worked in groups and created our website together, things changed. Through receiving peer feedback, I started to examine my own learning. This actually feels really good, and I even anticipate their feedback before receiving it.

Utilizing the "review-track changes" functions in Word documents in the curation process, students can follow peer comments and conduct the timestamped revisions. As Emily noted: *"Each person has a unique colour when commenting ... and I made my suggestions carefully because I must be responsible for my words."* Students also mentioned their *"iterative reflection and revision"* of their writing, WordPress websites and PowerPoint Slides based on peer feedback and AI-generated suggestions to *"match the visuals with the meanings"* (FG1-S2, Ella, James). For example, FG1-S2 shared her experience in curating her multimodal artefacts, *"at first, I just wrote the description and added pictures (see Figure 2). Then my groupmate suggested adding images that align with the description. So, I provided my description to AI and it gave me suggestions on image choice and colour scheme."* The final output of FG1-S2 (Figure 3) shows that the colour scheme and visuals match that of the themes for each block. She utilized dark pink for azalea blossom, grey for cave scene, and faded green for ancient villages. Also, in the cave scene description, she added a picture that specifically referred to "a stalactite like a helmet" in the descriptive text.

There are many places in Bijie which are worth traveling to. But the first thing to say is the BailiDuJuan Scenic Spot. It is located in the central area of the Bijie. It is the only azalea forest park in the world. It has a national azalea variety. If you like enjoying flowers, you can choose the place.



Zhijin Cave Scenic Area also is a famous scenic spot of Bijie. It's located in Zhijin, a city of Bijie. It's the largest karst cave in Asia. Many movies and TV are filmed there. Thousands of different forms stalactites in the cave. You can enjoy nature's uncanny workmanship, such as a stalactite like a helmet, it's 14 meter tall, so calls "overlord helmet". This stalactites after decades of formation. The constant temperature is around 15 degree Celsius in the cave. So, you can travel in summer.



Figure 2. Screenshot of FG1-S2's initial descriptive writing with images (excerpted).

Another example comes from Ella's group. After finishing the first draft, they were "not satisfied with the visual arrangement" because the dark navy-blue theme colours (see [Figure 4](#)) seemed "too rigid and serious". They planned to "utilise a commercial PPT template" yet had disagreement with the colour scheme. So, they asked AI for suggestions. The AI recommended using "low-saturation colours" and gave examples like "faded azure, misty seagrass, hazy lavender". They discussed and evaluated and finally chose a "faded-blue-beige" colour scheme (see [Figure 5](#)) because, as Ella said, "blue makes people feel calm and beige creates a warm feeling like stepping in the sand. We want to create a relaxing feeling and tell the audience it's a leisure journey to the warm South." She also added keywords, in the top right of the slide, to inform audience the highlights of the trip, which were also in perfect coherence with the four sites shown in the slide.

This iterative curation aligned with the project's emphasis on multimodal intertextuality, where students learned to weave texts, visuals, and hypermedia into cohesive narratives. This process also mirrors the "design" to "re-design" cycle in multiliteracies



Figure 3. Screenshot of FG1-S2's final post in WordPress-based portfolio (excerpted).

Shamian Island

Shamian Island attracts a large number of tourists with its picturesque scenery. Here, we can see many European-style buildings and the former sites of banks from the Republic of China era.



Figure 4. Screenshot of one slide from the first draft in Ella's group portfolio.

development, where critical engagement with multiple feedback sources nurtures design thinking.

4.1.2. Augmented confidence in multimodal meaning-making

Students highlighted how peer collaboration, enhanced by AI support, amplified their confidence, alleviating writing anxiety and increasing willingness to experiment in the first place. Lily said in her narrative: "I have never tried this way [curating web-based e-portfolios], and at first, I felt both excited and intimidated. But when working with my



Figure 5. Screenshot of the same slide from the final version of Ella's group portfolio.

classmates, I can feel a sense of security compared with when I worked alone." In all the focus group interviews, students emphasized how peer collaboration has relieved their pressure and fostered a feedback-rich learning environment, in which peer and AI-generated feedback played a pivotal role. Members in the fourth focus group noted: *"When we have conflicts [in deciding whether one's feedback is acceptable], we always ask for another member to judge or ask AI whose view is more reasonable."* Students used GenAI tools as both feedback providers and moderators in peer collaboration, fostering a personalized and learner-centred environment.

Students also demonstrated a growth in learner agency and positive affects because of the formative gradeless assessment and multisource feedback of portfolio compilation compared with the high-stakes examinations. As Emily reflected:

In exams, a red cross feels like a denial or punishment, and I'm always upset when seeing it. But this time, with review mode on, my friends' comments and teacher's comments are all presented in a less stressful way that triggers my curiosity and my willingness to revise, and sometimes we just start the conversation [in the dorm] about each other's writing. Also, AI provides feedback in a gentle way and even asks for my opinion on its feedback. I can feel being respected, so I'm quite enjoying these rewarding conversations.

The dialogic feedback from peers, teachers, and GenAI tools is leveraged in portfolio curation to expedite student feedback uptake, which considerably enhances students' engagement with multimodal artefacts.

More importantly, not only did students gain agency in writing and peer collaboration, but they also became more confident in multimodal knowledge construction, supported by AI as a multimodal design consultant. Students from both focus groups 1 and 3 mentioned the shift from textual writing to the coherent presentation of multimodal artefacts. For example, one student stated that:

At first, I thought my job was just writing these essays. But when we work with a WordPress website, I feel like a designer. Every colour, font, and image on our website is chosen to match our touring theme, which is a colourful culture tour [to Guizhou Province]. (FG1-S1)

Another student wrote in her reflective entry: “With images and other visuals like the headings or quotations, I can highlight my key points, and these colourful visuals make my writing attractive to my readers. (Student 12)” She also mentioned a design technique she learned from the AI. To “avoid visual fatigue”, people are supposed not to use overly bright colours and instead use shades from the same colour family as the main picture, which will create a more harmonious look. The multimodal analysis of student 12’s post cross-referenced her perceptions and application of this technique (see Figure 6). In her post, we see that the main picture features green, blue, and grey. She applied blue for quotations and faded blue with yellow for the content block. These matching colours provide harmonious visual effects and facilitate her multimodal description of this natural



Unique Huangguoshu Waterfall

*A thousand feet of shadow in front
of the waterfall wind, suspected to
flood the Milky Way.*

It is widely known that Journey to the West has a famous scene, which is where Tang Monk and his four fellows are walking on the waterfall. Well, I believe you have already guessed that it is the largest waterfall in the China-- Huangguoshu Waterfall.

Figure 6. Screenshot of student 12’s post on WordPress-based portfolio (excerpted).

scenery. When it comes to the content, she used a famous line by poet Li Bai (originally describing Huangguoshu Waterfall) in a quotation block as a hook. This shows her ability to enhance meaning-making from different perspectives and the effective use of cultural references.

The clearly outlined self and peer assessment rubrics are also a catalyst for students' development of confidence and metacognitive strategies in meaning-making. In the reflective entries, a majority of students mentioned the instructional effects of the rubrics when they assessed and gave feedback on others' work, highlighting that they *"always checked the assessment list item by item"*. Almost half of the class applied and reflected upon the scoring rubrics to assess their work when creating and curating the artefacts, and some of them *"provided the rubrics to AI"* and expected more detailed and rubric-referenced feedback. Though not directly, the succinct assessment rubrics enhance students' confidence in multimodal meaning-making by providing reference and guided instruction, and this positive effect will be further augmented with GenAI integration in content-specific, rubric-referenced feedback provision.

4.2. Expanded usefulness of e-portfolios for multimodal communication

4.2.1. Enhanced multimodal competence

In creating web-based portfolios, students explored functionalities and managed the WordPress platform with website guides and hands-on practices. This enhanced competence in the mastery of digital platforms is reflected in Mike's narrative:

It was difficult to build a website at first with unfamiliar interfaces, but with teacher demonstration and website guidelines, we figured out what to do after some practice. Anyway, it is very simple to undo the action or modify the content.

Students gradually increased their awareness of the "multisituational" and "multiform" nature of multiliteracies practices through creating and curating web-based portfolios. In reflective entries, students mentioned the global reach of their multimodal artefacts that *"people around the world who have access to the Internet can possibly view our websites"*, which is echoed by all the narrative inquiries highlighting the potential overseas audience of the websites and how they curated artefacts in audience-aware contexts.

The e-portfolio platform and the final group presentation also allowed and encouraged students to integrate various semiotic resources in conveying meaning. Emily reflected in her narrative on how she applied textual and multimodal visuals in presentation to enhance the delivery of meaning: *"Using both texts and images in presentation slides helped classmates understand better, especially when they weren't familiar with certain vocabulary"*. The first focus group used background music in their presentation to *"create a relaxing atmosphere"* because they *"wanted to express that this was a leisure trip."* In spatial forms, such as the design of the website layout, students adjusted the default website design and make it more *"reader-friendly"* with *"distinct division of sections and touring suggestions on food, accommodation, transportation and overall planning"* (FG1-S3). These enhanced multimodal competencies demonstrate a holistic multiliteracies developmental trajectory, from constrained, single-mode practices to creative, multimodal knowledge construction in an interactive digital space.

4.2.2. Critical evaluation of multisource feedback

Students developed a discerning approach to critically evaluate feedback from peers, instructors, and AI. When receiving peer feedback, they selectively integrated ideas rather than adopting them altogether. As students noted, *"I'd pick feedback that made sense to me. If something was unclear or I didn't agree with, I'd talk to my peers or group members to figure it out"* (FG3). Similarly, teacher feedback was treated as a guide rather than a directive. Students sought clarifications proactively: *"If I didn't understand the teacher's comments, I'd ask after class or ask AI to help explain the feedback in simpler terms"* (FG2-S4). This feedback uptake strategy, which blends human and AI mediation, mirrors a dialogic feedback process, where learners actively negotiate meanings across sources.

Metacognition also emerged as students reflected on feedback from multiple sources and compared it with their assessment. Many journal entries detailed "aha moments" when students recognized discrepancies between their self-assessment and feedback from teachers and AI. For example, a student from focus group 3 initially rated her essay as *"satisfactory"*, and her peer also gave her essay favourable feedback (see Figure 7). However, after comparing it with the teacher (see Figure 8) and AI-generated feedback, she realized that her argument lacks a clear logical flow and should be situated in certain cultural contexts (FG3-S1). For instance, the concept of "red spirit" may not be fully understood by visitors from countries outside China. She was then aware that her assessment of the writing was primarily anchored on vocabulary, grammar, overlooking the importance of effective intercultural communication and logical delivery. As a result,

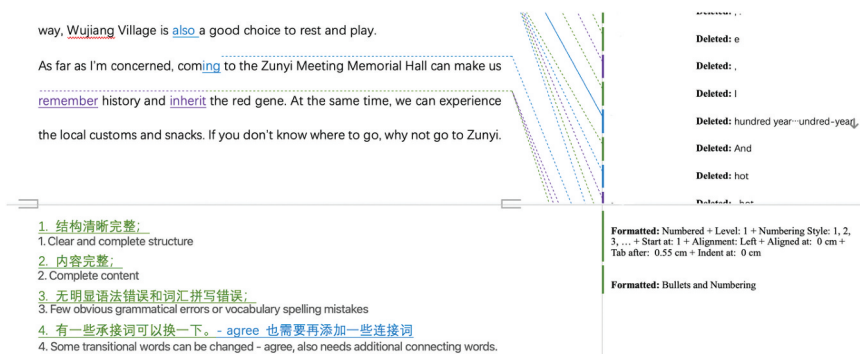


Figure 7. Peer assessment of FG3-S1's essay.

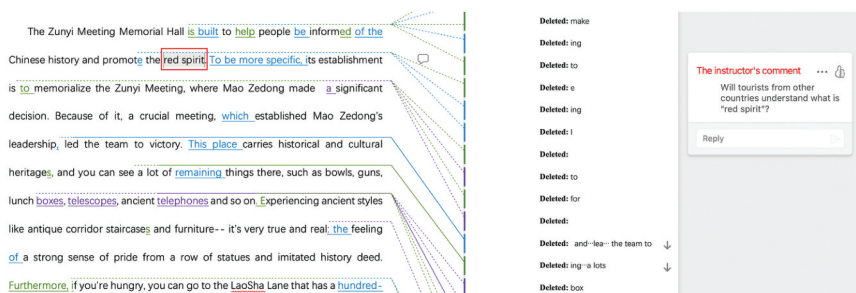


Figure 8. Teacher feedback on FG3-S1's essay.

she made revisions based on multisource feedback, with the help of AI as both feedback provider and for feedback synthesis. In her WordPress post (see [Figure 9](#)), the formerly “red spirit” was revised to “revolutionary spirit” for better delivery of meaning instead of mere translation word by word.

However, with AI-generated feedback, students frequently viewed AI comments as authoritative, for example, Lucy illustrated: “*I’d unconsciously trust it ... if AI suggested edits, I’d usually accept them.*” Fortunately, some students still maintain critical stances over AI-generated feedback. Emily disapproved of GenAI’s excessively intricate edits: “AI



The Zunyi Meeting Memorial Hall is built to help people be informed of the Chinese history and promote the revolutionary spirit of PRC (The people’s Republic of China) and fighters. To be more specific, its establishment is to memorialize the Zunyi Meeting, where Mao Zedong made a significant decision. Because of it, a crucial meeting, which established Mao Zedong’s leadership, led the team to victory. This place carries historical and cultural heritages, and there are a lot of remaining things, such as bowls, guns, lunch boxes, telescopes, ancient telephones

Figure 9. Screenshot of FG3-S1’s post on WordPress-based portfolio (excerpted).

occasionally transformed my basic sentences into intricate clauses, but I preferred maintaining my original style". To exert control, some improved their prompts, saying, "I learned to tell AI my English level and exactly what I needed. (FG4-S1)" "I would modify my questions till they made sense if the answer wasn't useful (FG3-S3)." This iterative negotiation portrays the developmental cyber-social literacy, indicating the scenario where learners scaffold AI's capabilities with their communicative intent. By juxtaposing human and AI-generated feedback, students cultivated a metacognitive awareness of how to learn, not just what to revise, which is a cornerstone of self-regulated multiliteracies development.

4.2.3. Readiness for real-world multimodal communication

The usefulness of e-portfolios, when empowered by GenAI tools, extends beyond mere language and literacy development to facilitating students' readiness for future careers in a multimodal world, transforming them from passive to proactive users seeking more useful tools to cater to their meaning-making needs.

Students increasingly reflected on their e-portfolio experiences and connected to real-world career demands. James highlighted practical benchmarks for his major: *"Tourism Management majors need strong English. Employers want CET-4 or CET-6 certificates and video editing skills. This project is a good start for me to navigate more skills I need to learn."* Mike also shared his vision of working in a popular industry nowadays, self-media on social media platforms. He noted, *"After graduation, I want to work in content creation for self-media. It requires not just writing skills but also multimodal aesthetics to attract audiences."* These reflections indicate a growing awareness of how multiliteracies align with employer expectations in a technology-driven job market.

Beyond textual interactions with GenAI, some students explored AI image generators to enhance their portfolios. For instance, one participant used AI tools to create visuals that *"matched [their] text perfectly and made it easier for readers to understand"* (Student 40). Albeit scarce among students, this proactive experimentation with multimodal synergy demonstrates their semiotic awareness in professional communication. As students bridged the gap between classroom learning and career readiness, they recognized the importance of adaptive skills. *"Nowadays, employers don't just want essays, they want pictures, infographics, videos, and websites. This project taught me how to learn up-to-date skills and deliver the message more effectively,"* noted Mike.

These findings demonstrate how students developed not only technical competencies but also critical awareness and strategic approaches to multiliteracies learning. However, challenges exist in their engagement with GenAI-assisted assessment.

4.3. Challenges in GenAI-assisted formative assessment

4.3.1. Initial resistance and entrenched habits

Students initially demonstrated resistance to utilizing AI tools in learning because many of them were only familiar with using search engines like Baidu as their primary information source and had no prior experience with GenAI tools. As Student 7 noted in his reflective journal, *"There are too many AI tools, and I don't know which one to choose or how to start with, or what to say to them."* However, after participating in an AI workshop in class, hands-on guidance helped students take their first steps. They gradually realized that *"different prompts can significantly influence the results"* (Lily) and appreciated that *"AI tools*

have no advertisements and provide more direct answers to my questions compared to search engines" (James). Nevertheless, during their interactions with AI, several students also mentioned the challenge of information overload. As one student (FG2-S3) responded, "AI provides too much information. Some answers were far beyond the sentence patterns I can understand and use." Emily also pointed out in her narrative that "getting satisfactory answers from AI requires multiple rounds of dialogue and negotiation."

Another concern raised because of the formative nature of e-portfolios. As first-year university students transitioning from exam-oriented education backgrounds, some struggled with the portfolio assessment tasks. They experienced feelings of discomfort, confusion, and even resistance because they were used to completing practice questions repeatedly and counting on memorization and had "a feeling of bewilderment" once they saw "tasks without model answers" (Student 39). Some students felt that building a website was "beyond their capabilities and unachievable" (Student 11). However, after the teacher's classroom demonstrations of website setup, task breakdown, and specification of timeline, students completed their creations and curated artefacts through guided instructions. Upon seeing their final results, students felt "proud of accomplishing something impressive" (FG4-S1). They also "realised that this approach of taking control and planning my own learning and writing content was more engaging than previous exams" (Lucy).

4.3.2. Dynamics with GenAI tools

Many students felt insecure when comparing their abilities to GenAI's advanced and almost instantaneous outputs. Mike described his English skills as "belittled by GenAI's advanced responses," while Student 15 questioned, "If AI can do it all, why bother learning English myself?" These reactions revealed a worrying belief that GenAI's skills diminished their worth. Additionally, although some students demonstrated an evaluative stance over AI-generated feedback, many students initially treated AI-generated feedback as unquestionably "authoritative" (Lily). As one participant admitted, "I treated AI's suggestions like teacher feedback. I just accepted everything" (Student 19). This uncritical acceptance, although easing revision workloads, is likely to erode students' agency in meaning-making processes and replace students' voices.

To address this, the teacher emphasized, in the AI workshop, that students should use GenAI as support while preserving the creativity and "human touch." Students were taught to ask AI "targeted questions" (FG1-S5) rather than input passive and over-generalized commands, a way that stimulates evaluative thinking skills. By positioning GenAI as a collaborator rather than a competitor, students can balance technological advances with the "human touch", forming an equitable human-AI partnership.

5. Discussion and implications

Drawing on first-hand evidence, this study provides insights into how students develop multiliteracies through portfolio compilation in GenAI-assisted portfolio assessment and how they perceive the role of GenAI in the process.

For **Research Question 1**, the study suggests that students develop their multiliteracies through three interconnected stages: initial transformation from passive to proactive learning supported by AI-empowered curation, growing confidence with enhanced

multimodal awareness, metacognitive development and professional growth. Portfolio assessment, unlike its high-stakes counterparts, emphasizes students' process of learning over the product of learning and creates a safety net for students with diverse language proficiency since they are assessed by their levels of engagement and commitment throughout the process (Lam, 2024a). During portfolio compilation, students gradually undergo an identity shift from passive receivers of knowledge to active learners who construct knowledge with multimodal semiotics, including not only visual but also embodied (Lim, 2020) and sensory (Mills, 2016) aspects. This enhanced learner agency (Lam, 2024a) is derived from the situated practices (New London Group, 1996) of this multiliteracies pedagogy that engage students with novel real-life tasks.

Students engaged in iterative cycles of creating, reviewing, revising, and "redesigning" (Cope & Kalantzis, 2015) multimodal artefacts based on multisource feedback, which gradually enhanced their authorship (Lam & Gan, 2024). This process of "limitless multimodal design" (Ho, 2024, p. 328) mirrors the design to re-design cycle fundamental to multiliteracies development (Cope & Kalantzis, 2023), where critical engagement with multiple feedback sources nurtures design thinking. GenAI played an auxiliary role in support of teacher feedback, namely as a feedback provider, a moderator in peer reviews, and a teaching assistant during the revising and reflection process. Students are immersed in the compilation process, which establishes not only a feedback-rich environment (Yancey, 2019) for active feedback uptake and follow-up actions (Carless & Boud, 2018) but also dialogic feedback that enhances the effectiveness of feedback as social communication (Carless, 2016).

The visual evidence from students' multimodal artefacts demonstrates that they evolved from simple text-with-image compositions to sophisticated multimodal ensembles (Hafner, 2014) where colour schemes, spatial arrangements, and semiotic resources were purposefully orchestrated. This development also aligns with what Kress (2010) terms "semiotic harmony", where different modes are coherently orchestrated and each contributes to a unified meaning. As applied by Hafner and Ho (2020) to digital multimodal composing, this harmony emerges through careful consideration of both multimodal affordances and orchestration. For instance, students' strategic use of colour schemes to match thematic content (such as the faded blue-beige palette for a "relaxing journey to the warm South") demonstrates their growing awareness of how different modes work together to create coherent meaning. This development of "semiotic awareness" (Towndrow & Pereira, 2018), the ability to select appropriate modes for specific purposes, is central to digital storytelling. Through compilation, students developed from focusing solely on text composing to understanding multimodal composition, viewing themselves as "designers" rather than just writers. To cater to their meaning-making needs, students learned to purposefully select colours, fonts, and images to match their themes and became more proficient and confident designers, integrating diverse semiotic resources. This aligns with the concept of "apt selection" (Lim et al., 2022), where students choose the most effective semiotic modes for specific communicative purposes. They also developed multimodal metalanguage, which suggests their rise of multimodal awareness and facilitates their feedback-giving practices (Liu & Lam, 2025).

Metacognitive development emerged as students engaged with multisource feedback, facilitated by the co-constructed assessment rubrics (Yan & Carless, 2022). The rubrics served multiple purposes: guiding self and peer assessment, promoting self-reflection on

multimodal compositions, providing reference for students' dialogic feedback practice with GenAI tools. That said, GenAI also plays an indispensable role in helping students alleviate writing anxiety and creating a more supportive learning environment (Dai & Liu, 2024; Meniado et al., 2024). The combination of peer, teacher and AI-generated feedback fosters a personalized, learner-centred environment. Students reflect on, compare, and critically evaluate multiple feedback sources, gradually developing their metacognition and evaluative thinking skills. This way, they developed cyber-social literacy (Kalantzis & Cope, 2025) by learning to effectively scaffold GenAI's capabilities with their own communication needs. More importantly, students' multiliteracies development brought by portfolio compilation extends beyond classroom context to connections with real-world professional demands as they began to explore additional GenAI tools proactively for enhanced multimodal creation, not just for the assigned assessment tasks but for the development of workplace competencies. This expansion of skill sets reflects what Xia et al. (2024) identify as crucial for balancing AI use with autonomous learning capacity. It is also what formative assessment calls for, to equip students with learning-how-to-learn skills (Lam, 2024a).

For **Research Question 2**, students perceived both significant affordances and notable challenges with the integration of GenAI into portfolio assessment. On the empowering side, GenAI served as a versatile learning companion by offering gentle, constructive feedback (Hopfenbeck et al., 2023), moderating peer discussions, while helping students understand complex teacher feedback. Their critical evaluative skills are cultivated when they reflect on, compare, critically evaluate AI-generated feedback and customize their prompts for better outputs (Ma et al., 2024). Additionally, the AI-assisted portfolio compilation prepared students to communicate and make meaning in an increasingly multimodal world and helped with their goal-setting and action-taking for career-relevant learning needs and future professional lives. However, GenAI integration also poses challenges to students' learning. Many students initially struggled with resistance and technical barriers because of their unfamiliarity with AI technology and their preference for traditional search engines. They also faced difficulties managing information overload (Lam & Gan, 2024) and complex AI responses that sometimes exceeded their language proficiency and were valueless for reference. Worse still, the integration of GenAI, though mostly restricted to feedback provision, created power dynamics where students either felt "belittled" because of their composing skills diminished by GenAI's almighty capabilities, or initially accepted AI-generated feedback uncritically as an authority.

Therefore, three key pedagogical implications emerge. First, teachers should carefully plan multimodal portfolio tasks that draw on students' shared experiences to enhance motivation (Cope & Kalantzis, 2023), while providing sufficient scaffolding for their multiliteracies development in the portfolio compilation process. Timely guidance and instruction should be provided to students along their multimodal meaning-making process. The design and assessment should also incorporate formative feedback with the elements of metacognitive reflection, and co-constructed assessment criteria to promote students' feedback literacy and form a positive learning environment (Yan & Carless, 2022).

Second, although GenAI provides valuable support for portfolio assessment, teachers still need to scaffold and guide students to overcome initial barriers and develop positive synergy with GenAI tools. The study suggests that students' initial resistance and anxiety are effectively addressed through an AI workshop that provides structured guidance to

help them understand how to start and communicate effectively with AI tools while maintaining their voices. Furthermore, the use of reflective questions (i.e. to what extent you disagree with AI comments) as part of the portfolio task can facilitate students' development of metacognitive and evaluative skills.

At last, teachers and universities should collaboratively construct a learning environment that encourages multimodal experimentation and multiliteracies growth. This includes creating a low-stakes, feedback-rich environment that fosters peer collaboration and dialogue. Teachers should also focus on building students' confidence in multimodal meaning-making while supporting their transition from traditional exam-oriented learning approaches to learner-centric, process-orientated learning.

6. Conclusion

This study investigated students' developmental trajectories of multiliteracies through GenAI-assisted portfolio assessments and explored their perceptions of GenAI assistance in this innovative pedagogy. The findings reveal a transformative journey in students' multiliteracies development, characterized by increased engagement, expanded multimodal competencies, and enhanced metacognitive awareness. Throughout portfolio compilation, students evolved from passive recipients of knowledge to active designers of multimodal meaning and demonstrated increasing confidence and proficiency in the use of a variety of semiotic resources.

The study makes important contributions to the portfolio assessment scholarship. First, it provides empirical evidence for the effectiveness of integrating GenAI tools into portfolio assessments as a pedagogy of multiliteracies. The visual documentation of students' multimodal artefacts offers concrete evidence of how peer, teacher, and AI-generated feedback can synergize to create a feedback-rich learning environment that supports students' multimodal knowledge construction and self-regulated learning. Second, it extends our understanding of "cyber-social literacy learning" in authentic teaching practices, demonstrating how students gradually learn to collaborate with GenAI tools in feedback practices while maintaining critical awareness. The study illuminates how students navigate the relationship between human creativity and AI support, and how they develop strategies to leverage AI capabilities while preserving their creative insights. Third, as students increasingly connect their multimodal practices to real-world career needs, the findings highlight the potential of GenAI-assisted portfolio assessments to bridge the gap between language classrooms and students' future professional lives.

However, limitations exist in this study. The participants of this research are first-year undergraduate students who have just finished the Chinese College Entrance Examination and have little knowledge about advanced GenAI technology. Therefore, the effects of the AI workshop and guidance on their portfolio compilation are salient. Future research can continue to examine the effectiveness of this multiliteracies pedagogy in different educational contexts. Also, researchers can implement this pedagogy in senior year classes where students have different levels of AI literacy and examine whether it can still help cultivate multiliteracies and cyber-social literacy.

In the study, GenAI was used primarily for dialogic feedback and assessment practices, with little application in multimodal image creation and visual design. Future research could explore the integration of GenAI tools into these creative perspectives.

Furthermore, GenAI-assisted portfolio assessment through the medium of social media deserves investigation and research (Lam, 2024a), given that social media platforms have become important spaces for learning and communication among a new generation of language learners in this GenAI era.

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Author contributions

CRedit: **Yue Ni**: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Ricky Lam**: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing.

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Appendix. Co-constructed rubrics for self and peer assessment

评价内容 Assessment Content	标准描述 Standards	优秀 Good	一般 About right	不足 Insuffi- cient	具体反馈说明 Qualitative feedback
总体结构 Overall Structure	分段清晰, 每段都有明确的功能和目的 Clear introduction, body paragraphs, and conclusion with logical flow				
内容 Content	有标题 Title: Clear and engaging 具体地标及地点类型 Location and landmarks: Specific and helpful 游客可选择的吸引物 Type of attraction: Natural scenery, cultural heritage, etc. 游客可选择的活动 Activities: What visitors can do/eat/experience/see/feel 推荐理由及解释说明 Reasons for recommendation and explanation 出行选择及其原因 Touring choices: Transport, accommodation, budget, etc.				
语法 Language	作文中语法准确, 表意清晰 Grammatically accurate with few errors Clear delivery of message				
词汇 Lexis	作文中词汇使用准确、多样且符合语境 Diverse and appropriate vocabulary choices Uses topic-specific terms				
视觉呈现 Visual Presentation	图片与内容匹配 Images match content 色彩协调且易于浏览 Coherent colour scheme and easy to navigate 使用多模态元素增强意义表达 Use multimodal elements to enhance meaning 创造性使用WordPress功能及PowerPoint设计元素 Creative use of WordPress features and PowerPoint design elements				
总体评价 Overall Assessment					